Today's agenda

- Intro
- Types of graph
- Storage
- BFS (level order) +1

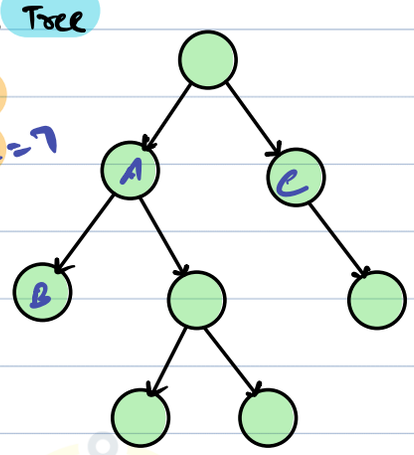
AlgoPrep

Intro

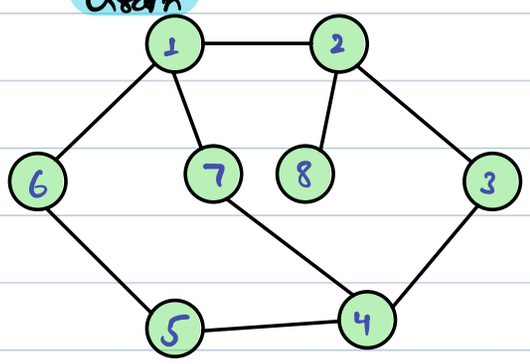
↳ Graph: Connection of nodes and edges

1. Tree

N = 8
E = N-1 = 7



Graph



N = 8
E = 9

1 and 4 are nbrs of 7.

→ diff^n bet^n trees and graphs.

1. Nodes in graph can have any number of connections without hierarchy.

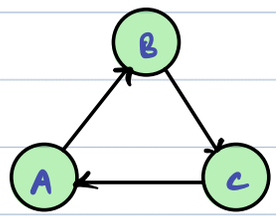
↓

↳ Graphs can have cycles.

2. Any directional movement is allowed in graph.

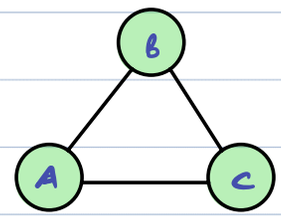
* Classification of graphs

Case I: Based on types of edges.



↳ directed graph

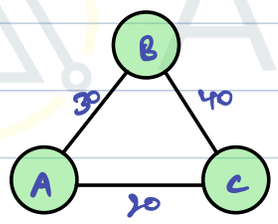
↳ insta followers



↳ undirected graph

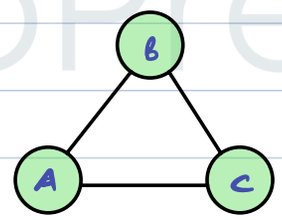
↳ facebook friend

Case II: Based on weight of edge



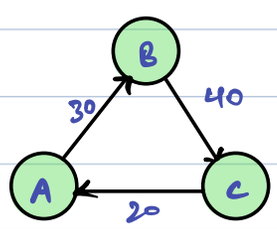
↳ weighted graph

↳ google maps



↳ unweighted graph

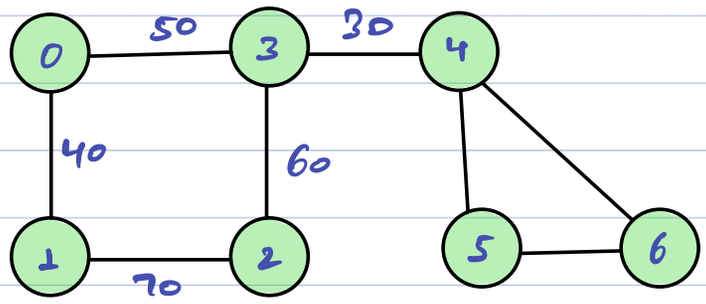
↳ ??



→ directed weighted graph

Storing a graph

undirected graph



input

N = 7 → nodes

M = 8 → no. of edges

3	4	30
1	2	70
2	3	⋮
4	6	⋮
0	1	⋮
4	5	⋮
5	6	⋮
0	3	⋮

① Adjacency matrix representation

Graph

nodes

	0	1	2	3	4	5	6
0	0	~~0~~ 1	0	~~0~~ 1	0	0	0
1	1 ~~0~~	0	~~0~~ 1	0	0	0	0
2	0	1 ~~0~~	0	3 ~~0~~	0	0	0
3	~~0~~ 1	0	1 ~~0~~	0	30 ~~0~~ 1	0	0
4	0	0	0	1 ~~0~~ 30	0	~~0~~ 1	~~0~~ 1
5	0	0	0	0	~~0~~ 1	0	~~0~~ 6
6	0	0	0	0	1 ~~0~~	~~0~~ 1	0

7×7

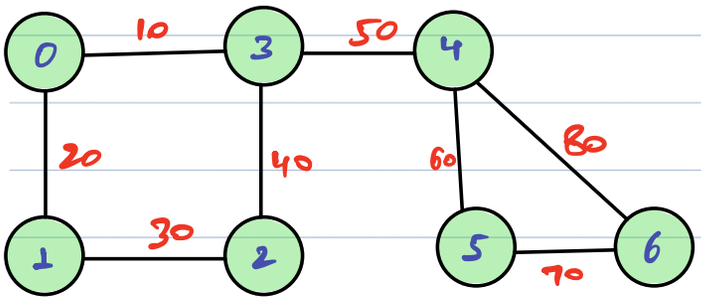
0 → disconnected

1 → Connected

issue:

↳ Space wastage

② adjacency list representation



N=7 M=8

edges

3	4
1	2
2	3
4	6
0	1
4	5
5	6
0	3

```
List<List<Integer>> graph = new ArrayList<>();
for(int i=0; i<n; i++) {
    graph.add(new ArrayList<>());
}
```

0	1 3
1	2 0
2	1 3
3	4 2 0
4	3 6 5
5	4 6
6	4 5

graph

//Pseudo Code

```
Class Pair {
    int v;
    int wt;
}

construction (int n, int m, int edges[m][2]){

    List<List<~~Integer~~ Pair>> graph = new ArrayList<>();
    for(int i=0; i<n; i++) {
        graph.add(new ArrayList<>());
    }

    for(int i=0; i<m; i++){
        int u = edges[i][0];
        int v = edges[i][1];
        graph.get(u).add(v);
        graph.get(v).add(u);
    }
    return graph;
}
```

AlgoPrep

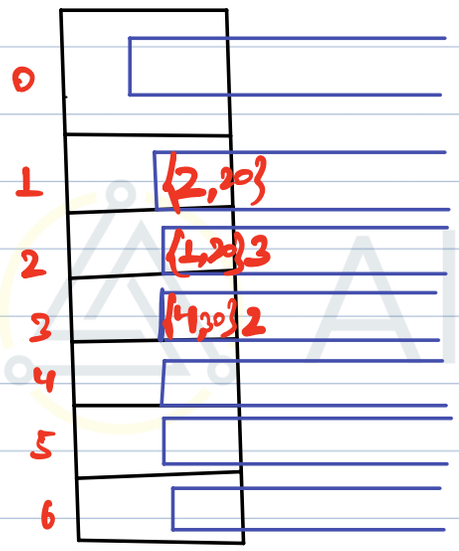
```
construction (int n, int m, int edges[m][2]){
    List<List<Integer>> graph = new ArrayList<>();
    for(int i=0; i<n; i++) {
        graph.add(new ArrayList<>());
    }

    for(int i=0; i<m; i++){
        int u = edges[i][0];
        int v = edges[i][1];
        graph.get(u).add(v);
        graph.get(v).add(u);
    }
}
```

u = 2
v = 3

N = 7 M = 8

edges

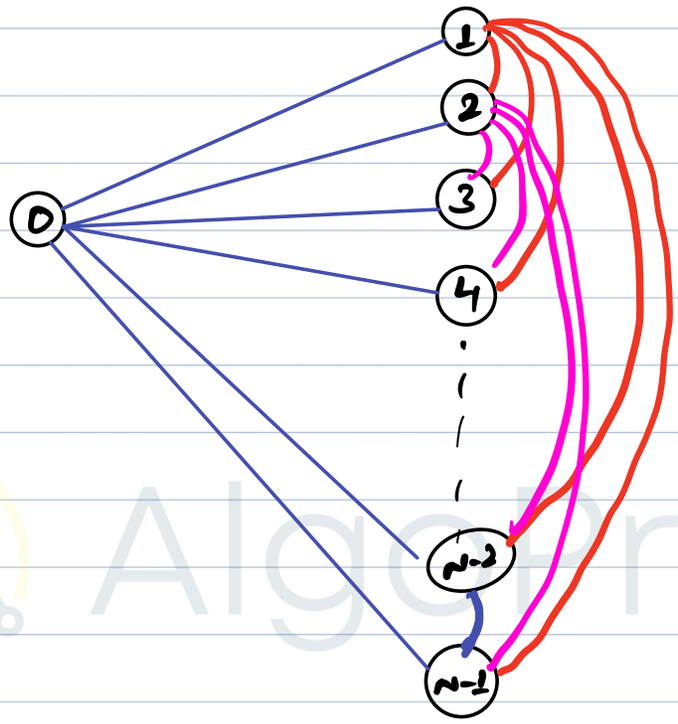
	0	1	
0	3	4	30
1	1	2	20
i → 2	2	3	
3	4	6	
4	0	1	
5	4	5	
6	5	6	
7	0	3	



graph

AlgoPrep

Q) Find max no. of edges possible if N nodes are present in Graph.



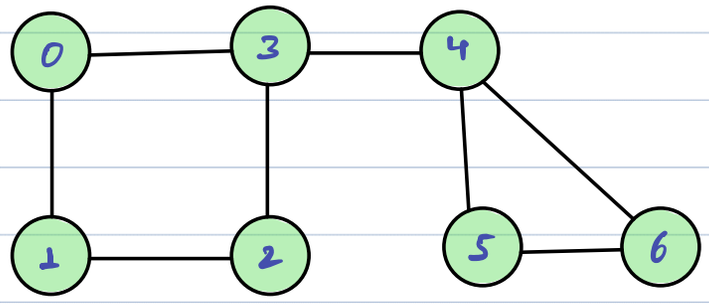
$$(N-1) + (N-2) + (N-3) + \ldots ___ ___ + 1$$

$$\frac{N*(N+1)}{2} \qquad \frac{(N-1)*N}{2} \simeq \frac{N*(N-1)}{2} \approx O(N^2)$$

Break till 9:50 PM

→ Breadth first search

BFS traversal



i=0

N=7 → No. of nodes

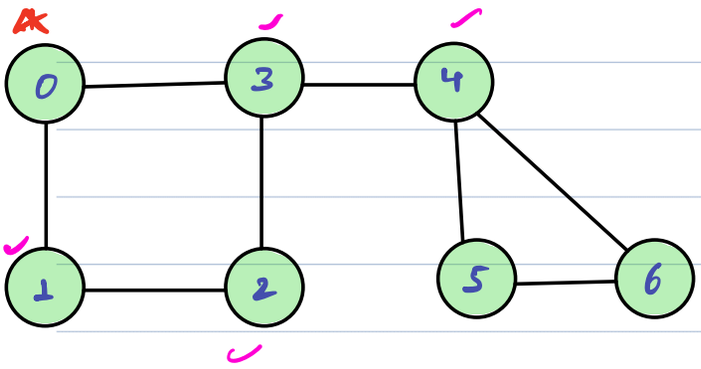
M=8 → No. of edges

	0	1
0	3	4
1	1	2
2	2	3
3	4	6
4	0	1
	4	5
	5	6
	0	3

```
List<List<Integer>> graph = new ArrayList<>();
```

graph

0	1	2	3	4	5	6
1, 3	2, 0	1, 3	4, 2, 0	3, 6, 5	4, 6	4, 5

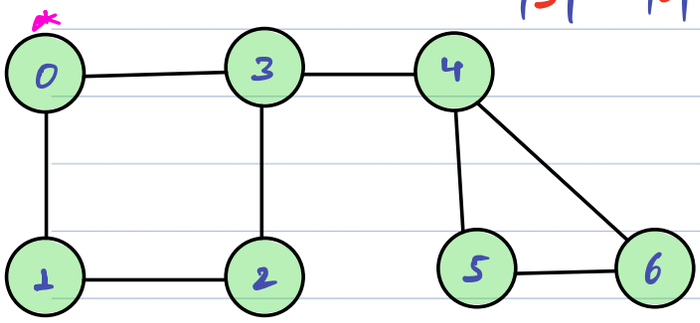


→ you will have to decide the start point for bfs.

0 (1 3) (2 4) (5 6)

graph

0	1	2	3	4	5	6
1, 3	2, 0	1, 3	4, 2, 0	3, 6, 5	4, 6	4, 5



queue : ~~0 1 3 2 4 6 5~~

vis :

0	1	2	3	4	5	6
~~F~~T	~~F~~T	~~F~~T	T~~F~~	T~~F~~	T~~F~~	~~F~~T

rem = 5

0 (1 3)(2 4)(6 5)

0 (1 3) (2 4) (5 6)

AlgoPrep

//Pseudo code

```
void BFS (int N, int M, int[][] edges){

    List<List<Integer>> graph = Construction(N,M,edges);

        Queue<Integer> q;
        boolean[] vis = new int[N];

            q.add(0);
            vis[0] = true;

T.C: ??     while (q.size() > 0){
                int rem = q.remove();
                S.O.P(rem);

                //add all unvisited nbrs.
                List<Integer> nbrs = graph.get(rem);

                for (int v: nbrs){
                    if (vis[v] == false){
                        q.add(v);
                        vis[v] = true;
                    }
                }
            }
    }
```